JIM DAOU
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CPSC 6420 ASSIGNMENT 2

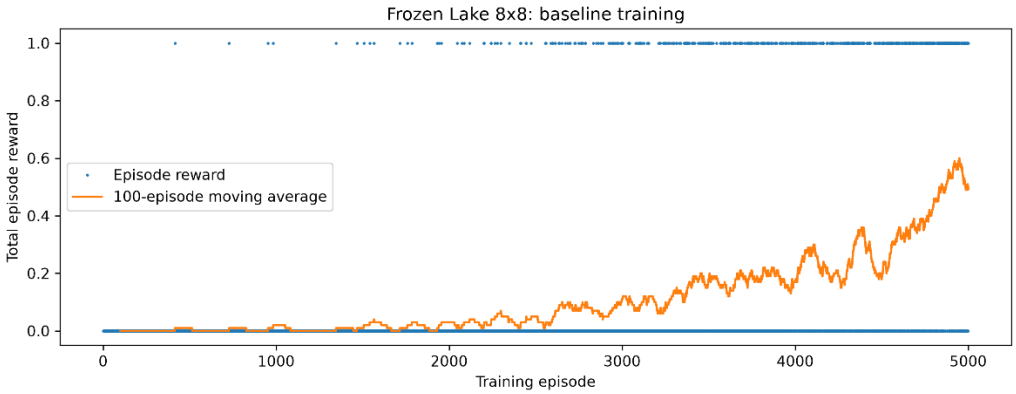
(A) **Environment description**. Describe the environment's goal, state/observation space, action space, transition dynamics, reward structure, and termination conditions. Explain any state discretization or environment settings you use.

a. The environment goal is for the agent to reach the goal(G) from the start(S) while avoiding tiles that have holes(H)
b. Observation space is 64 in the 8x8 map
c. Action space is 4; agent can move up, down, right, and left
d. Transition dynamics of frozen lake depend on if we have defined 'is_slippery'. With it set to 'True', the default behavior causes the agent to only have a 1/3 chance of moving in its intended direction and 1/3 chance of moving in either direction perpendicular to its intended direction. With it set to 'False', the agent always moves in its intended direction.
e. Reward structure for the agent is +1 for reaching the goal and 0 for everything else, including moving to a tile with a hole
f. Termination happens when the agent moves to a tile with a hole or reaches the goal tile. Episode is truncated at 200 steps for the 8x8 environment map.
g. No further state discretization settings or environment settings were used
h. The observations stated here were gathered using 'inspect_frozen_lake.py' and official Frozen Lake documentation

(B) 'agent.py' was created and features an agent that uses Q-Learning

(C) Baseline configuration chosen has epsilon = 1, gamma = 1, alpha = 0.1, and 5000 episodes

a. Epsilon decays at a rate of ({number of episodes} – {number of episodes observed}) / {number of episodes). This can be considered our exploration schedule.
b. This is the plot of a baseline training session. We observed a rolling average that increased overall, but was rough and had lots of ups and downs



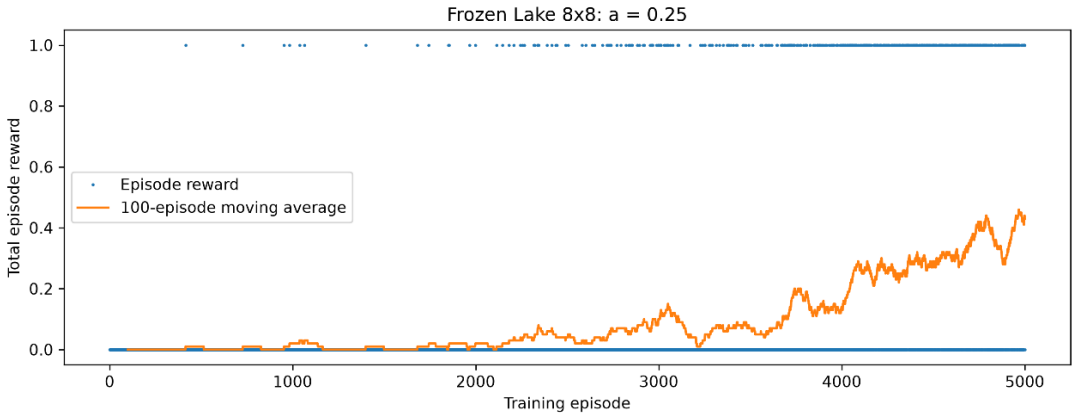
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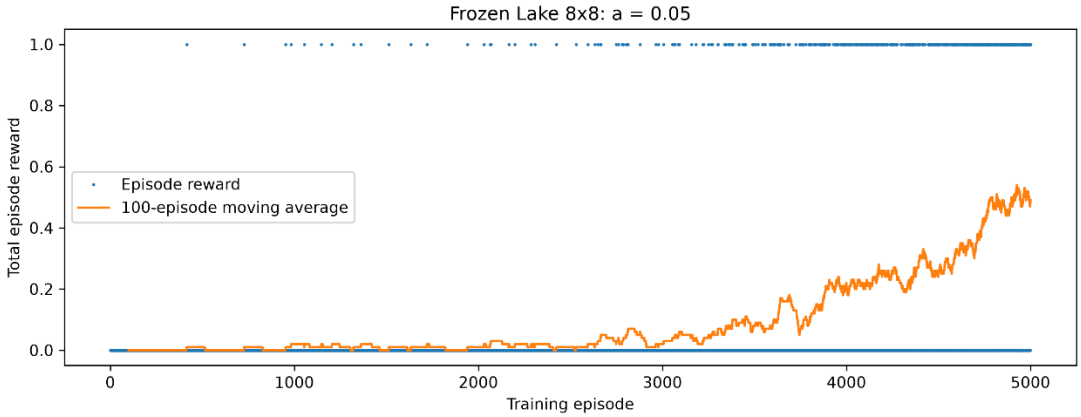
c. This behavior could be attributed to a suboptimal learning rate or the randomness of Frozen Lake's 'is_slippery' feature

(D) Learning rate tests were conducted with alpha 0.25, 0.05, and 0.01. Baseline was used for all other variables.

a. At alpha 0.25, we notice a similar learning speed to the baseline; however, there is less stability and slightly worse overall performance. Higher learning rate did not improve the agent.



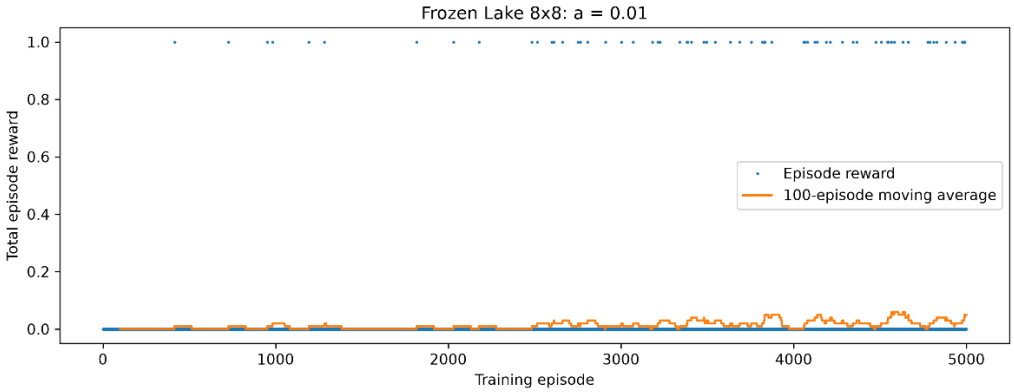
b. At alpha 0.05, there is a noticeably slower learning rate compared to baseline, but stability is slightly smoother. Still, overall performance is about the same.



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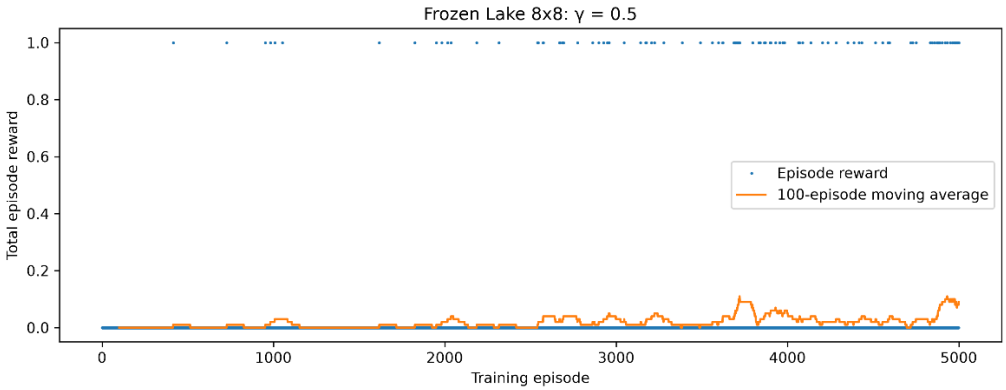
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c. At alpha 0.01, the agent fails to meaningfully learn. There is nearly no observed learning rate, and the overall performance reflects this. An alpha this low likely hinders the agent from meaningfully incorporating new observations.



(E) Discount factor tests were conducted with gamma 0.5, 0.9, and 0.99. Baseline was used for all other variables.

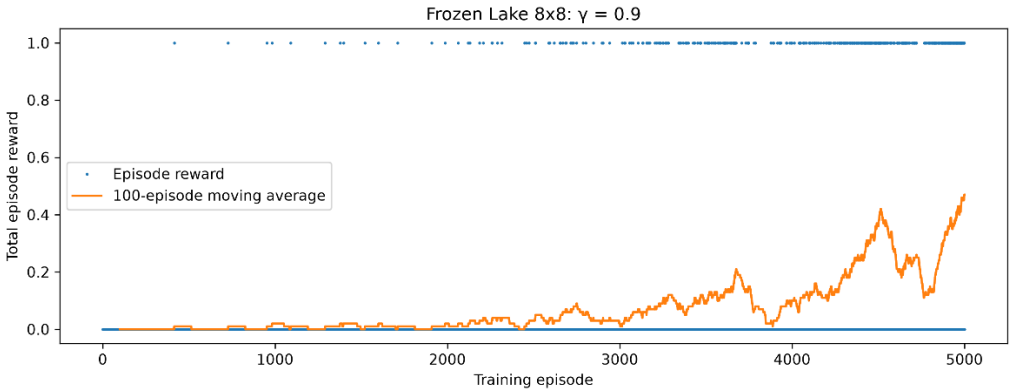
a. At gamma 0.5, there is very slow growth, if any at all. Heavy discounts for delayed rewards do not help this agent, likely due to there being only 1 reward and that reward only appearing after many steps.



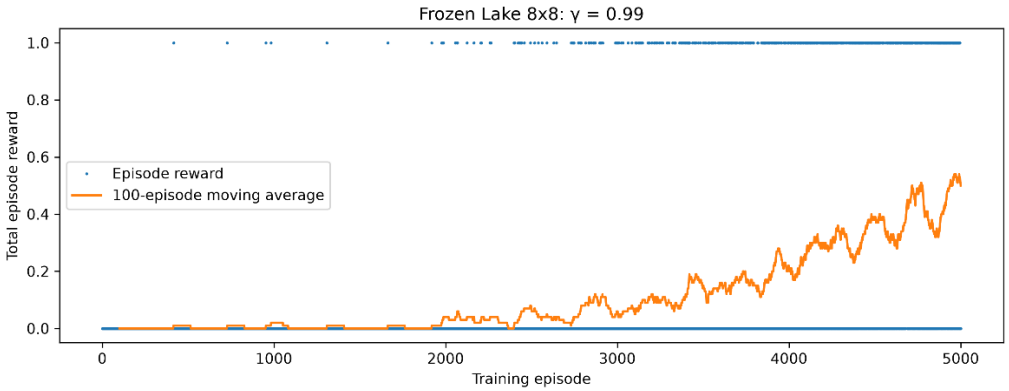
b. At gamma 0.9, there initially is accelerated growth to maximum performance, but then there is a steep drop in learning rate followed by a steep increase. Since the peak of average reward over 100 episodes is achieved sooner, this gamma could be considered to positively affect the agent.

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c. At gamma 0.99, learning rate has the most confident positive trend, despite subpar stability. Still, compared to the baseline, this agent starts learning noticeably around episode 2000, while baseline starts learning noticeably around episode 1500. Further, overall performance for this agent is still slightly worse compared to baseline. Overall, discounts on delayed rewards do not seem to help the agent.

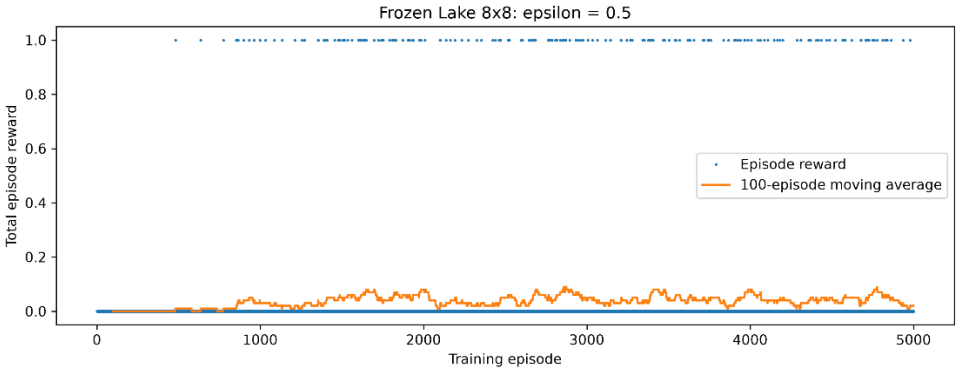


(F) For the additional hyperparameter, we chose to update epsilon and its decay with static value 0.5, 0.5 with baseline decay, and baseline value with a different decay method.

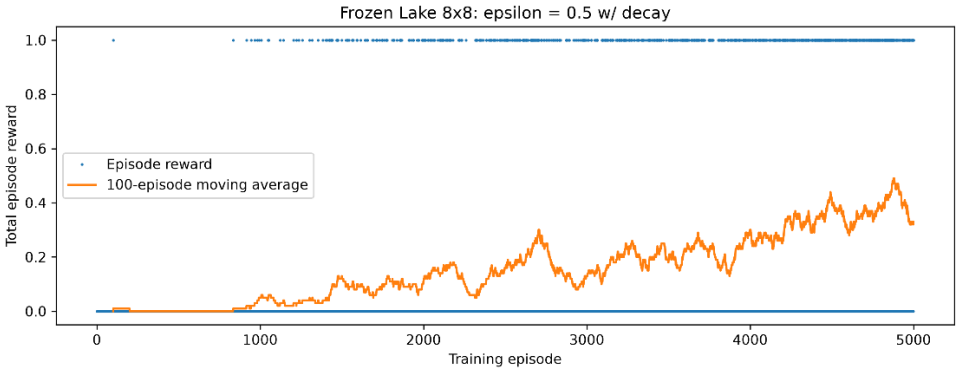
a. At epsilon with static value 0.5, there is minimal learning observed with the agent. The agent has a 50/50 chance to exploit; however, with the random factor of Frozen Lake, there is an even less chance that the agent actually takes an intended action, which hinders learning.

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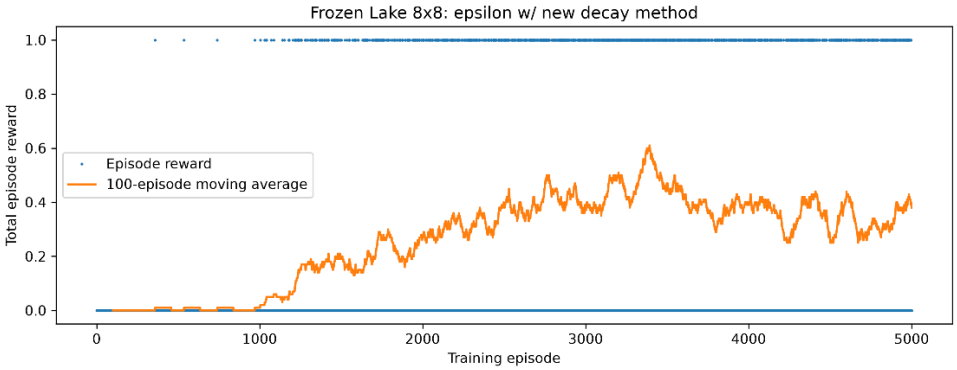
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b. At epsilon with value 0.5 and baseline decay, learning rate improves dramatically compared to the agent with a static epsilon. Further, compared to baseline, this agent starts learning sooner, at episode ~900 compared to ~1500



c. At baseline epsilon with different decay method, (0.999 ** {number of observed episodes}), the learning rate of this agent takes on a new shape not seen before. Learning rate increases dramatically after episode 1000 and reaches its peak around episode 3,500, proceeding to curve down afterwards. While this agent beats the baseline agent to peak learning rate, it fails to keep peak learning rate, and baseline is the better agent at the end.



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(G) The best agent observed thus far has been the baseline agent. Many agents came close to being better but ultimately failed in some meaningful way. The baseline agent shows the most stability, commendable learning rate after a certain number of episodes, and best overall performance after all training episodes.

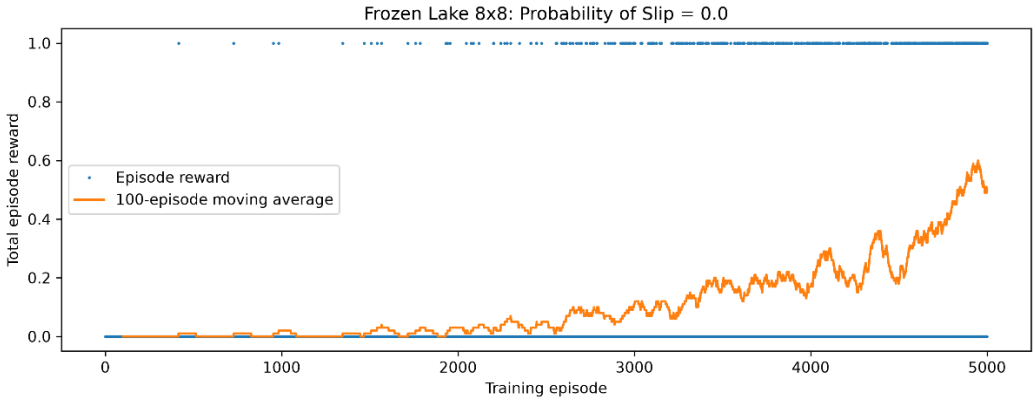
a. Under 500 evaluation steps at epsilon 0, the best agent- baseline agent- has a 51.6% success rate with each episode having an average of 83 steps

```
Evaluating the agent's performance over 500 episodes...
Average reward over 5000 training episodes: 0.112
Average reward over 500 evaluation episodes: 0.516
Average steps over 500 evaluation episodes: 83.15

(CPSC-6420-Q_Learning_Agent) C:\Users\Porta-Rex\Documents\CPSC-6420-Q_Learning_Agent>
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(H) To introduce another source of stochasticity, we introduced another slip factor that randomly overrode the current action with the previous action. Controlling this factor was the variable 'probability_of_slip', which we tested at 0, 0.1, 0.25, and 0.5

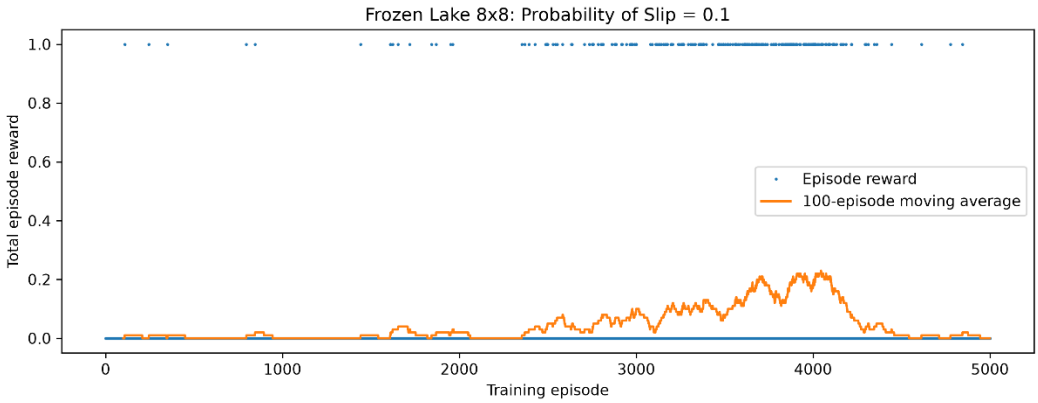
a. At value 0, the training and evaluation episodes were the same as what was observed with the baseline agent. This was only the case after making sure the script only called np.random() if there was a value greater than 0 for the variable first as random numbers were still being consumed in checking if the additional slip factor took place.



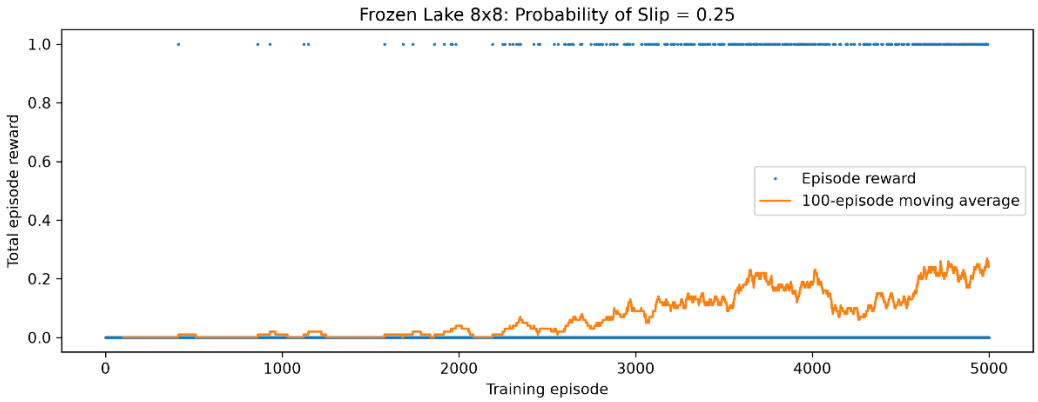
b. At value 0.1, the learning rate and evaluation outcome were hindered greatly. There was still some semblance of learning, but the original policy was subpar at 10% noise.

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c. At value 0.25, the learning rate and reward improve, especially at the final episodes; however, the rate is still very unstable and is nowhere near the robustness of the original policy before noise



d. At value 0.5, stability improves slightly and total episode reward increases slightly but is still incomparable with the original policy. Thus, it is observed that the original, baseline policy does not handle noise well.

